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MAT 524: Functions of a Real Variable II
Homework V

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$\Big[\Big[$ Problem One $\Big]\Big]$:

Problem Statement

Let V be a normed vector space over $\mathbb{R}$. For a linear functional $T : V \to \mathbb{R}$, show that the following are equivalent.

(a) T is bounded

(b) T is continuous

(c) T is continuous at $\vec{0}$.

Solution

$\{$(a) $\Rightarrow$ (b)$\}$: Suppose T is bounded. Then there exists a finite real number $M \geq 0$ such that for all $v \in V$,

$$|Tv| \leq M \|v\| . \tag{1}$$

Let $\varepsilon > 0$. Choose $\delta = \frac{\varepsilon}{M} > 0$. Whenever $v, w \in V$ and $\|v - w\| < \delta$, we have

$$\begin{aligned} |Tv - Tw| &= |T(v - w)| && (2) \\ &\leq M \|v - w\| && (3) \\ &< M\delta && (4) \\ &= \frac{M\varepsilon}{M} && (5) \\ &= \varepsilon. && (6) \end{aligned}$$

Therefore, T is continuous.

{(b) $\Rightarrow$ (a)}: **(Based on the lecture notes)** Suppose T is continuous. Let $\varepsilon = 1$. Then there exists $\delta > 0$ such that, for all $v \in V$, if $\|v\| \leq \delta$, then $|Tv| < 1$. For nonzero v, consider the vector $w = \frac{\delta}{\|v\|}v$. Since $\|w\| = \delta$, it follows (by continuity) that

$$
\begin{aligned}
|Tw| &= \left|T\left(\frac{\delta}{\|v\|}v\right)\right| && (7)\\
&= \frac{\delta}{\|v\|}|Tv| && (8)\\
&< 1. && (9)
\end{aligned}
$$

Rearranging (9) yields

$$
|Tv| < \frac{1}{\delta}\|v\|. \tag{10}
$$

Thus, T is bounded.

{(b) $\Rightarrow$ (c)}: If T is continuous, then it is continuous at $\vec{0}$.

{(c) $\Rightarrow$ (b)}: Suppose T is continuous at $\vec{0}$. Let $\varepsilon > 0$. There exists $\delta > 0$ such that for all $v \in V$, if $\left\|v - \vec{0}\right\| = \|v\| < \delta$, then $|Tv - T\vec{0}| = |Tv| < \varepsilon$. Let $x, y \in V$. Then $x - y \in V$, and, if $\left\|x - y - \vec{0}\right\| = \|x - y\| < \delta$, then $|T(x - y) - T\vec{0}| = |Tx - Ty| < \varepsilon$. Therefore, T is continuous.

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$\Big[\big[$ Problem Two $\big]\Big]$:

Problem Statement

Let V be a normed vector space over $\mathbb{R}$ and let V^* be its dual (the set of bounded linear functionals on V) with the operator norm

$$\|T\| = \inf\{M : |Tv| \leq M \|v\| \text{ for all } v \in V\}.$$

(a) Show that the above is indeed a norm.

(b) Show that $\|T\| = \sup_{\|v\|=1} |Tv|$.

(c) Show that V^* is complete (so it is a Banach space).

Solution

$\blacksquare$

[[Problem Three]]:

Problem Statement

Let V be a normed vector space. Show that the closure of any subspace of V is a subspace.

Solution

Let W be a subspace of V, and let $\overline{W}$ be the closure of W. First note that $W \subseteq \overline{W}$, meaning anything that is an element of W is also an element of $\overline{W}$. Furthermore, $\overline{W}$ is nonempty, since W is nonempty.

Now we show that $\overline{W}$ is closed under addition. Let $x, y \in \overline{W}$. Suppose $x, y \in W$. Then $x + y \in W$. Hence $x + y \in \overline{W}$. Next, suppose (without loss of generality) that $x \in W$ and $y \in \overline{W} \setminus W$. Let $\varepsilon > 0$. Since y is a limit point of W, there exists $w \in W$ such that $\|w - y\| < \varepsilon$. But we can rewrite this inequality as

$$\begin{aligned} \|w - y\| &= \|w - y + x - x\| && (11) \\ &= \|w + x - (x + y)\| && (12) \\ &< \varepsilon. && (13) \end{aligned}$$

Since $w + x \in W$, it follows that $x + y$ is a limit point of W, i.e., $x + y \in \overline{W}$. Finally, suppose that $x, y \in \overline{W} \setminus W$. Let $\varepsilon > 0$. There exist $w_1, w_2 \in W$ such that $\|w_1 - x\| < \varepsilon/2$ and $\|w_2 - y\| < \varepsilon/2$. Thus,

$$\begin{aligned} \|w_1 + w_2 - (x + y)\| &= \|w_1 - x + w_2 - y\| && (14) \\ &\leq \|w_1 - x\| + \|w_2 - y\| && (15) \\ &< \varepsilon/2 + \varepsilon/2 && (16) \\ &= \varepsilon. && (17) \end{aligned}$$

Since $w_1 + w_2 \in W$, this implies that $x + y \in \overline{W}$.

Now we show that $\overline{W}$ is closed under scalar multiplication. Let $\lambda \in \mathbb{C}$ and let $x \in \overline{W}$. Let $\varepsilon > 0$. Then there exists $w \in W$ such that $\|w - x\| < \varepsilon / |\lambda|$. Thus

$$\begin{aligned} \|\lambda w - \lambda x\| &= |\lambda| \, \|w - x\| && (18) \\ &< |\lambda| \, (\varepsilon / |\lambda|) && (19) \\ &= \varepsilon. && (20) \end{aligned}$$

Since $\lambda w \in W$, it follows that $\lambda x \in \overline{W}$.

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